

1) Firmware Library

A firmware library is a collection of prewritten functions, drivers, & reusable code used in embedded system programming.

It helps developers interact with hardware devices such as GPIO Pins, UART, SPI, sensors, display & communication modules without writing everything from scratch.

It enables developers to reduce development time, & improve code readability.

eg: ~~gpio~~ ~~library~~ 1) `gpio_set (pin_no, state)`

2) `HAL` library

3) `Arduino` library etc

2) Why APIs are important in embedded system?

An API (Application Programming Interface) is a set of functions & rules that allow software to communicate with hardware or other software modules.

Instead of direct modifying hardware registers, programmer can call API functions.

`GPIO_set (pin_no)`

APIs provide a simple way to access peripheral such as,

* `GPIO`

* `UART`

* `ADC`

+ `Timer`

* `sensors`

+ `WiFi/Bluetooth` modules.

Importance:

- * API enable hardware abstraction
- * code reusability
- * Faster development
- * Easier debug & portable

3)

What was understood from the lab code? ^{how}
From the lab code, the main understanding ^{was} ^{using}
embedded software interact with hardware ^{using}
APIs & library.

gpio.h :

In this header file all the macros &
API functions are written.

gpio.c :

In gpio.c all the necessary library &
function definition are written.

main.c :

In main.c all the library & dependency
files are included & written an software style
embedded code for configuring a (PORT).